

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

Quiz no. 1

Summer Semester 2021-2022

DURATION: 30 MIN

FULL MARKS: 15

Math 4641: Numerical Methods

Figures in the right margin indicate marks.

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| 1. | A football is kicked by a player while it is 0.5 meters above ground. Based on the trajectory of the ball, the height h of the ball is modeled by the following nonlinear equation, where t is in seconds:
$h(t) = -4.9t^2 + 7.7t + 0.5$
After how many seconds will the ball reach the ground again? Find the result using Newton-Raphson method guaranteeing at least 3 significant digits correct. | 9
(CO2)
(PO1) |
| 2. | Provide a comparative analysis between Secant method and False Position method. | 6
(CO3)
(PO2) |